

OpenLoop

The AI case agent for bank files that stay open for weeks

Code enforces the rules · Claude plans the next step · people make the decisions

NEXXT AI Hackathon 2026 · Raiffeisen Bank Romania · AI for Operational Efficiency · Petru & Teodor

Back-office work is an open loop

Documents arrive over days; every one must agree with the others and with the policy

95 days

and about $2,598 per corporate KYC review (2023 vendor survey)

30 days

a trade-register certificate stays valid, then the file must ask for a new one

14 days

to report a beneficial-owner mismatch (AMLR art. 24, from July 2027)

Today a person chases every missing paper, **re-checks** every expired certificate and **reconciles** owners and addresses by hand.

“Procesarea desfăşurându-se manual” — Raiffeisen Bank Romania on a 14,000-client remediation (research/01)

OpenLoop in one picture

One append-only, hash-chained ledger per file — every actor reads it and writes to it

Events in

documents · emails · portal uploads · staff actions · time passing

→

Code pipeline

extraction with verbatim quotes · registry lookup · hard rules HR-01…08 · timers and expiry

→

Claude plans

Opus 5.5 on Amazon Bedrock decides the next step over judgment-only tools

→

Gate

only actions the rules allow can run · approve credit is always DENIED

→

People decide

short decision cards · rulings become precedents and learned checks

Judge Featherless classifier + similarity check — auto-accept only when both agree

Two screens, one ledger the bank's work queue and the client portal, each in its own words

One workflow, input to outcome

Brutăria Aurora's SME loan file: the demo in the video

Input

a new loan case from the client's phone · the application pack · certificates · e-mails · days passing

Rules and checks

fields extracted with exact quotes · 16 rules: trade register, ANAF, beneficial owners, DSCR · two judges for fuzzy matches

Outcome or flag

clean documents pass in seconds · a wrong file is rejected with the reason · a conflict opens a decision card and the 14-day legal clock

Human review

two analyst clicks; the ruling becomes a precedent · file complete: the reviewer checks the ledger · the credit decision stays human

In the demo file a person makes 2 decisions; the agent does the rest. Synthetic data; registry and e-mail simulated; time jumps are simulated.

Agent design: a hybrid by evidence

Code where rules are certain, Claude where judgment is needed, people where it matters

Code always runs

extraction with verified quotes, registry, rules, timers, expiry — deterministic

Claude plans

next steps, email intent, escalation wording; 20 s budget per click

Fails safe

errors or timeouts fall back to the scripted planner; a code safety net runs after Claude

Gate on every action

allowed by the rules or denied and logged (GATE-01/02/03)

Humans in the loop

cards show only what differs; rulings become precedents

Learns safely

3 similar rulings propose a check; backtested; a different person approves

Measured live: 3.5–13 s per click, 0 fallbacks · every step logged as pipeline / claude / planner / safety-net

Tested, calibrated, measured

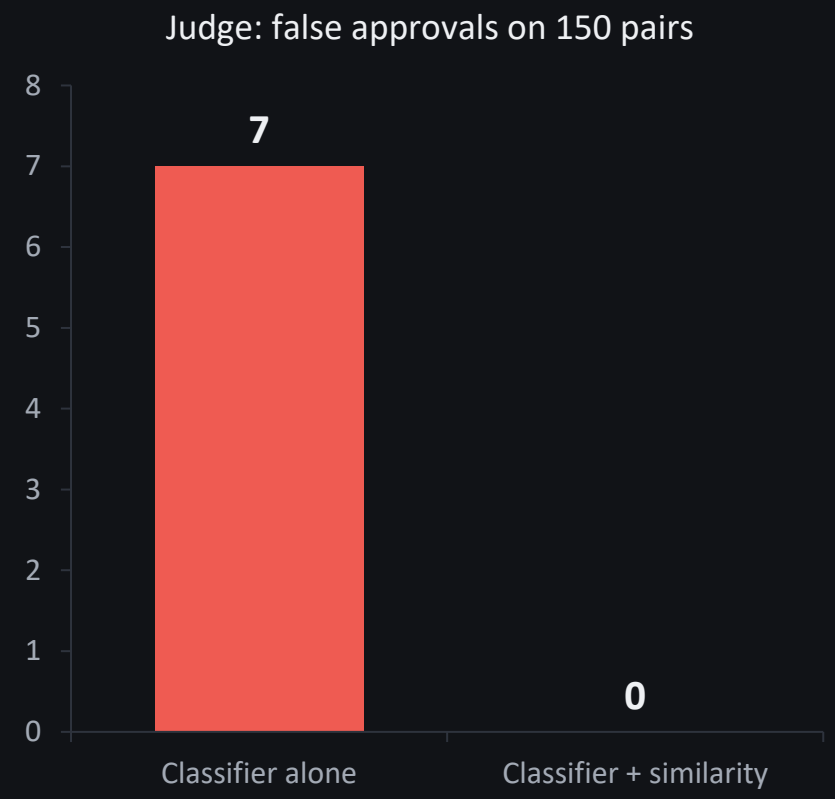
Evidence in the repo, reproducible offline

201

backend tests: the full demo, every rule path, role leaks, tamper detection

8 / 8

synthetic companies end on their intended problem (one rule path each)



8 → 0

false approvals of the learned address check after small-town counterexamples

≈ $0.26

of Claude calls per file (list prices); p50 52–60 ms with 200 users

Reproducible in one command

FastAPI · SQLite event store · React bank app and client portal · Claude on Bedrock · Featherless

```
pip install -r requirements.txt
cd web && npm ci && npm run build && cd ..
python -m uvicorn app.main:app --app-dir backend
# bank /   ·   client portal /client/
```

Offline mode (scripted planner, heuristic judge) runs the whole demo with no API keys. **Live mode** swaps in Claude and the Featherless judge with the same UI and the same ledger.

Deployed on Railway with sign-in and 2FA, four staff roles and a fail-closed access policy on every API route.

- **Data stays in the bank**

  one append-only table; PostgreSQL on-prem in production

- **Bring your own document AI**

  extractors are adapters; quotes are verified by code

- **Privacy by construction**

  client views are allow-lists; staff identities never reach clients

- **All synthetic data**

  documents follow real ONRC / ANAF / F20 forms; tax IDs in an unassigned range

Romanian rules, in code

Two templates on one engine: SME loan and KYB check · Annual business review

HR-01	CUI checksum
HR-02	Company active in the trade register
HR-03	Certificate under 30 days old
HR-04	Owners ≥25% match the register (Law 129/2019)
HR-05	An ID for every owner
HR-06	Same registered office everywhere
HR-07	Debt service coverage ≥ 1.2
HR-08	No tax debts on the ANAF certificate
SR-01	Staff contact change → credit action in 48 h
SR-02	Client disputes a staff change

Judge questions JQ-01 (same address?) and JQ-02 (signatory = administrator?) decide only after the hard facts match.

Where it pays off

The agent does the work between human decisions

100,000+

active Smart Business SME clients at Raiffeisen Romania (H1 2026)

1 or 5 yrs

AMLR periodic refresh by risk, from July 2027: every SME, again and again

Path to production

1 **Shadow mode**

annual business reviews with one back-office team

2 **SME onboarding**

the full loan and KYB template

3 **Same engine**

garnishments (2.25M ANAF acts in H1 2025), DORA incident clocks, complaint deadlines

Analysts see only files that need a decision · clean files finish on their own · every action is auditable

Why it's new

Time is first-class

the file is a ledger; rules run over sequences, not single documents

An agent that cannot overstep

a gate between plans and actions; credit decisions stay human

It learns from people, safely

rulings → precedents → backtested checks with maker-checker approval

One ledger, two audiences

the bank and the client see the same file, each in their own words

What's limited, and what's next

Limits today

- Synthetic data only; the trade register, ANAF and e-mail are simulated
- PDFs are stored but not read: a person reads them (no OCR yet)
- Credit-bureau consent is recorded, not yet enforced
- Our own security audit: 2 Medium findings still open
- Templates are code: the admin template editor is not built

Next: a 4-week shadow pilot

- **Where** annual business reviews with one back-office team
- **Plug in** the bank's document AI and the ONRC / ANAF feeds, on-prem
- **Measure** hours per file, days to complete, re-requests, audit findings
- **Decide** with numbers from the bank's own files